What is data structure?

A data structure isa way of organizing data so that it can be efficiently utilized.Almost all software systems and programs that have been built employ many basic and complex data structures. Therefore, we must be knowledgeable in data structures.

What is an algorithm?

Refers to a sequence of actions taken to complete a particular task or to get the intended result. Algorithms are clearly defined procedures that assist computers in problem-solving.It's like the building blocks that make it possible for computing devices to work well and make decisions. Each line of code that the program writes is a kind of algorithm that tells it what to do.

Relationship between data structure and algorithm

Algorithms have been used to manipulate thedata structure through different programming languages. On the other hand, data structure helps improve an algorithm's performance. Therefore algorithms and data structure depends on each other. For example, the algorithm can perform the search, insert, update, delete and sort operations on the data in the data structure

Why data structure and algorithm isessential?

Data structure and algorithms help us store and process data efficiently, considering time and memory usage, which improve the performance of computer programs and computer systems.

Data types

In computer science and computer programming, a data type (or simply type) is a set of possible values and allowed operations. A data type tells the compiler or interpreter how the programmer intends to use the data. The data type defines two important things.

1) Defines a certain domain of values (It determines which values can be stored)
2) Define the operation allowed on those values

Example:

For int type

- It takes only an integer value

- Operations: Addition, Subtraction, multiplication, division, bitwise operations etc.

For float type

- It takes only a floating value
- Operations: Addition, Subtraction, multiplication (bitwise operations and % are not allowed)

There are two types of data types:

- **Primitive data types:** A primitive data type specifies the size and type of variable values. These are data types built into a programming language and can be used to derive other data types (non-primitive data types). Unlike Non-primitive one's primitive data types store only one value. Examples of primitive data types are int, float, double etc.
- **Non-primitive data types (Abstract data types).** These are user-defined data types used to define the values of non-primitive data types. The user defines the values and operations manually. On the other hand, the system can also determine the operations using the built-in function. Abstract data types can store more than one value at the same time. Examples of Non-primitive data types are array, structure and union.

Since data types operations involve storing and manipulating data in the memory, they are also called data structures. There are two types of data structures:

- Primitive data structure
- Non-primitive data structure

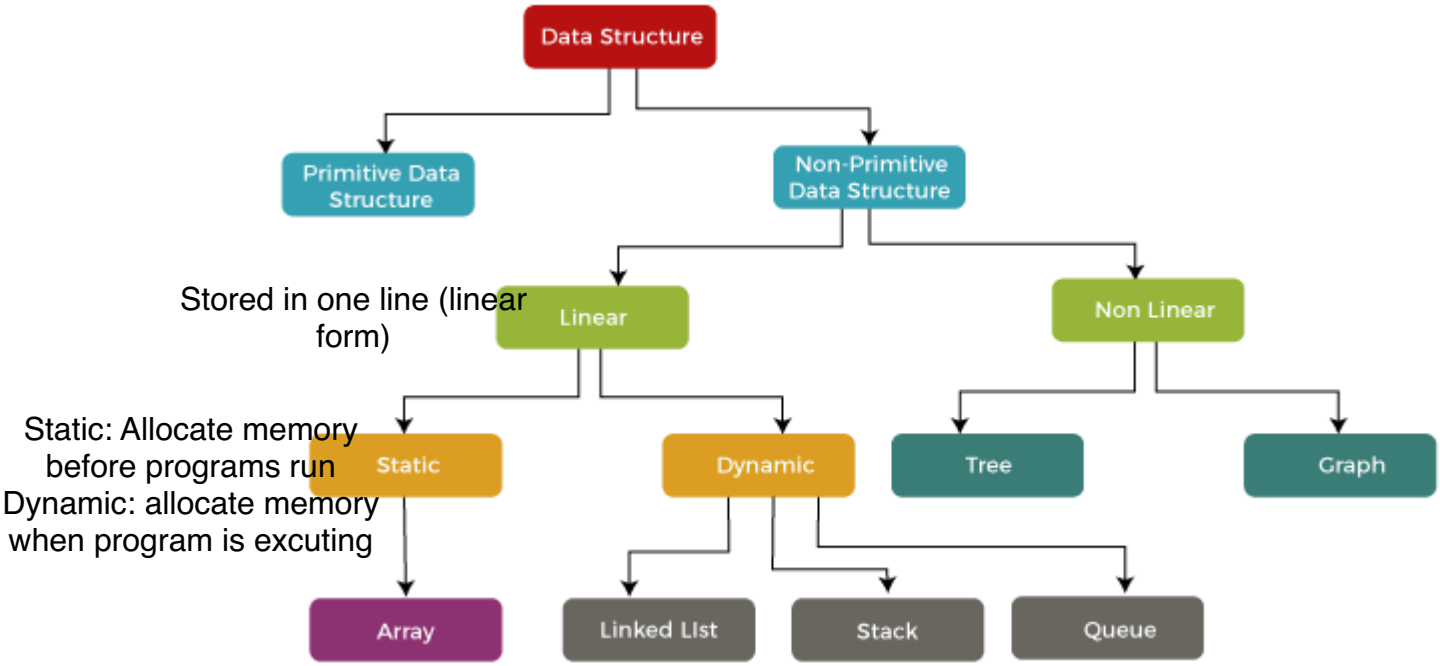


Figure 1: Classification of data structure

Primitive Data structure

The primitive data structures are primitive data types. The primitive data structure is a fundamental data structure that stores only one data simultaneously. The int, char, float, double, and pointer are primitive data structures that can hold a single value.

Non-Primitive Data structure

The non-primitive data structure is a type of data structure which is user-defined that stores the data of different types in a single entity.

- **Linear data structure**

  A linear data structure is one in which data items are organized sequentially or linearly, with each component connected to the elements before and after it. Examples of the linear data structure are array, stack, queue and linked list

- **Static data structure**

  The size of the structure in a static data structure is fixed. It is possible to change the data structure's content without impacting the amount of memory space assigned to store the data. An example of a static data structure is an array

- **Dynamic data structure**

  The size of a dynamic data structure is not fixed and may be changed while operations are being performed on it. Dynamic data structures are intended to allow for data structure changes during runtime. Example of dynamic data structure is Que, stack and linked list

➢ **Non-linear data structure**

Non-linear data structures are those in which data components are not arranged sequentially or linearly. Therefore, we cannot explore all of the items of a non-linear data structure in a single run. Examples of non-linear data structures are trees and graphs

Major Operations which can be performed in the data structure

The major or common operations that can be performed on the data structures are:

- **Searching:** We can search for any element in a data structure
- **Sorting:** We can sort the elements of a data structure either in an ascending or descending order
- **Insertion:** We can also insert the new element in a data structure.
- **Updation:** We can also update the element, i.e., we can replace the element with another element
- **Deletion:** We can also perform the delete operation to remove the element from the data structure

Functions/Methods

A Java method is a collection of statements grouped to perform a certain operation in a Java program. For example, a section of a bank information system dealing with customers' deposit may contain several lines of code that communicate with the database, which transfer and store the

desired amount in the customer's account. All program statements that perform deposits can be grouped into a function.

The syntax for creating a function

access_specifier static return_type method name (list of arguments)

```
{
    //Function Body
}
```

Example of function

```
public static float average (float marks)
{
   //Codes for calculating the average goes here
}
```

Definition of terms from the function above

public: An access specifier which indicates that the function can be accessed outside the class

static: Indicates that the **function** can be called without creating an **object.** Alternatively, we can say that no **object** owns this function.

float: It is the **return** type which indicates that our **function** will return average marks as a float value. Otherwise, it should be declared void if the function returns nothing.

average: It is a class name

marks: parameter variable (This part is optional)

Question: Write a Java program which implements a function concept that calculates the average of three input Marks, such as Mathematics, Geography, and English, and outputs the results on the screen.

```java
package BICT_DAY;
import java.util.Scanner;
public class Main {
    public static void evarage()
    {
        float math;
        float geog;
        float engl;
        Scanner input=new Scanner(System.in);
        System.out.println("Enter Mathematics Marks");
        math=input.nextFloat();
        System.out.println("Enter Geograph Marks");
        geog=input.nextFloat();
        System.out.println("Enter English Marks");
        engl=input.nextFloat();
        float average=(math+geog+engl)/3;
        System.out.println("Average Mraks is "+average);
    }
    public static void main(String[] args) {
        // TODO code application logic here
        evarage();
    }
}
```

Recursion

Recursion in java is a process in which a method calls itself continuously. A method in java that calls itself is called recursive method.

It makes the code compact but complex to understand.

Syntax

returntype methodname()

{

//code to be executed

methodname();//calling same method

}

Example: A program to display ‘hello’ infinite times

```java
package BICT_DAY;
public class Main
{
    static void p()
    {
        System.out.println("hello");
        p();
    }
    public static void main(String args[])
    {
        p();
    }
}
```

Example: A java program to calculate factorial of 5

```java
package BICT_DAY;
public class Main
{
    static int factorial(int n)
    {
        if(n==1)
            return 1;
        else
            return (n*factorial(n-1));

    }
    public static void main(String args[])
    {
        System.out.println("Factorial of 5 is"+factorial(5));
    }
}
```

Java Classes and Objects

What is a class in Java?

A Class is an object constructor or a "blueprint" for creating objects. Or Class is a group of variables and methods of different data types.

What is an object?

An object is any real-world entity which can be converted into a computer program. The object can be physical or logical (tangible and intangible). An example of an intangible object is the banking system. Examples of tangible objects include a chair, bike, marker, pen, table and car.

An object has three characteristics:

State: These are properties of an object. For example, car's object state will be color,current speed, number of wheels etc. In programming, object states are represented by variables.
Behaviour: These are all actions that an object can perform. For example, for a car, object behaviour will be slow down, speed up and turn around. In programming, objects' behaviours are represented by functions.
Identity: It is a unique name for an object which differentiates an object from the rest

Declaring an object

Classmate objectname=new Classname ();

Example of Class Student

```
Class Student
{
        //Instance variables
        Static/Non static Variable 1;
        Static/Non static Variable 2;

        Static/Non function 1(parameter1, parameter2…..parametern)
        {
          //Local variables
           Variable 1;
           Variable 2
        }
        Static/Non Function 2(parameter1, parameter2…..parametern)
```

```
        {
          //Local variables
           Variable 1;
           Variable 2

        }
        Public static void main (String args[])
        {
            //The declarations of object "c"
              Student  c=new Student();

        }
}
```

Note the following:

1) Declaring Objects is also called instantiating a class
2) **An instance variable**: Avariable which is created inside the class but outside the method and not declared as **static**. Instance variable doesn't get memory at compile time. It gets memory at runtime when an object or instance is created. That is why it is known as an instance variable.
   - **Characteristics of instance variables**
     a) Instance variable can be accessed directly by any non-static method. Static method cannot access instance variable accept the main method
     b) The main method can access the instance variables using an object
3) **Static variable:**Avariable which is created inside the class but outside the method and declared as **static.**
   - **Characteristics of static variables**
     a) Static variable can be accessed directly by any method regardless it is a static or not without an object
4) **Local variable:** Avariables which created within or inside a function block or function parameters.

- **Characteristics of local variables**

  a) These variables can only be accessed within the function in which they are declared. The lifetime of the local variable is within its function only, which means the variable exists till the function executes.

5) **Instance methods/functions:**Is a method that is declared in the class without using the static keyword.

   - **Characteristics of Instance Method**

   a) Instance method can only be accessed by the main function using objects

6) **Static methods/functions:**Is a method that is declared in the class using the static keyword.

   - **Characteristics of Static Method**

   a) Static method can be accessed by any function/Method including non static function directly without using object

Example: A Java program to print student name and subjects scores using object concept

```java
package BICT_DAY;

public class Main
{
    static String name="Onesno";
    public void modules()
    {
        int mathematics=90;
        int geography=40;
        System.out.println("Student Name is "+name);
        System.out.println("Mathematics Marks is "+mathematics);
        System.out.println("Geograph Marks  is"+geography);
    }


    public static void main(String args[])
    {
        Main xc=new Main();
        System.out.println("Student Name is "+name);
        System.out.println("...........Student Results are..........");
        xc.modules();
    }
}
```

Example: A Java program to print student name and subjects scores using object and passing of variables concept

```java
package BICT_DAY;

public class Main
{
    String name="Onesno";
    public void modules(int math,int geo)
    {
        int mathematics=math;
        int geography=geo;
        System.out.println("Student Name is "+name);
        System.out.println("Mathematics Marks is "+mathematics);
        System.out.println("Geograph Marks  is"+geography);
    }


    public static void main(String args[])
    {
        Main xc=new Main();
        System.out.println("Student Name is "+xc.name);
        System.out.println("..........Student Results are......... ");
        xc.modules(90,40);
    }
}
```

In real time development, we create classes and use it from another class. It is a better approach than previous one. Let's see a simple example, where we are having main () method in another class. We can have multiple classes in different Java files or single Java file. If you define multiple classes in a single Java source file, it is a good idea to save the file name with the class name which has main () method.

Example: A program implementing the main function in another class

```java
package BICT_DAY;

class studentResults{
    String name="Onesno";
    public void modules()
    {
        int mathematics=90;
        int geography=40;
        System.out.println("Student Name is "+name);
        System.out.println("Mathematics Marks is "+mathematics);
        System.out.println("Geograph Marks  is"+geography);
    }

}

public class Main
{
    public static void main(String args[])
    {
        studentResults xc=new studentResults();

        System.out.println("..........Student Results are......... ");
        xc.modules();
    }
}
```

Constructors

In Java, a constructor is a block of codes similar to the method. It is called when an instance of the class is created. At the time of calling constructor, memory for the object is allocated in the memory.

Rules for creating Java constructor

There are two rules defined for the constructor.

1. Constructor name must be the same as its class name
2. A Constructor must have no explicit return type
3. A Java constructor cannot be abstract, static, final, and synchronized

Example: A program to implement constructor without passing of parameters

```java
package BICT_DAY;

public class Main
{
    String name="Onesno";
    Main()
    {
        int mathematics=90;
        int geography=40;
        System.out.println("Student Name is "+name);
        System.out.println("Mathematics Marks is "+mathematics);
        System.out.println("Geograph Marks  is"+geography);
    }


    public static void main(String args[])
    {
        System.out.println("..........Student Results are......... ");
        Main xc=new Main();
    }
}
```

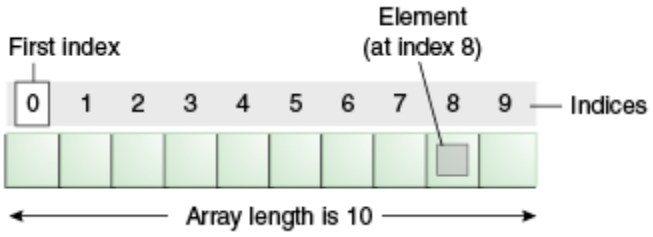
Example: A program to implement constructor with passing of parameters

```java
package BICT_DAY;

public class Main
{
    String name="Onesno";
    Main(int math, int geo)
    {
        int mathematics=math;
        int geography=geo;
        System.out.println("Student Name is "+name);
        System.out.println("Mathematics Marks is "+mathematics);
        System.out.println("Geograph Marks  is"+geography);
    }


    public static void main(String args[])
    {
        System.out.println("..........Student Results are......... ");
        Main xc=new Main(90,40);
    }
}
```

Array Data Structure

Java array is an object which contains elements of a similar data type. Additionally, the elements of an array are stored in a contiguous memory location. It is a data structure where we store similar elements. We can store only a fixed set of elements in a Java array. Array in Java is index-based, the first element of the array is stored at the 0th index, 2nd element is stored on 1st index and so on.

Consider a description of an array A[10] below;



Types of Arrays in Java

There are two types of arrays.

- Single Dimensional Array
- Multidimensional Array

i. Single Dimensional Array

In a single-dimension array, elements are arranged in linear order.

The syntax for declaring a single-dimension array

a) dataType[] arr= new dataType[Array size]; or
b) dataType []arr= new dataType[Array size]; or
c) dataType arr[]=new dataType[Array size] ;

Bubble Sort Algorithm Using Array

Bubble sort is a simple sorting algorithm. This sorting algorithm is comparison-based algorithm in which each pair of adjacent elements is compared and the elements are swapped if they are not in order.

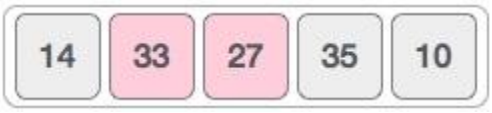
How the Bubble Sort Works?

An example on how to sort an array **int A[]=new int{14,33,27,35,10}** using bubble sort

Bubble sort starts with very first two elements, comparing them to check which one is greater.

In this case, value 33 is greater than 14, so it is already in sorted locations. Next, we compare 33 with 27.

We find that 27 is smaller than 33 and these two values must be swapped.



The new array should look like this −

Next we compare 33 and 35. We find that both are in already sorted positions.

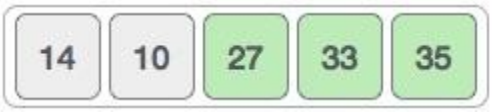
Then we move to the next two values, 35 and 10.

We know then that 10 is smaller 35. Hence they are not sorted.

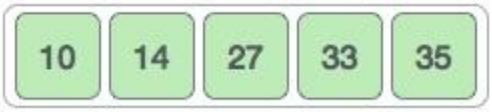
We swap these values. We find that we have reached the end of the array. After one iteration, the array should look like this −

To be precise, we are now showing how an array should look like after each iteration. After the second iteration, it should look like this −

Notice that after each iteration, at least one value moves at the end.



And when there's no swap required, bubble sorts learns that an array is completely sorted.



Now we should look into some practical aspects of bubble sort.

Bubble Sort Algorithm using array

1. Declare an Array ‘A’
2. Store elements in the Array
3. Declare variable K which store intermediate/temporary results when swapping array contents

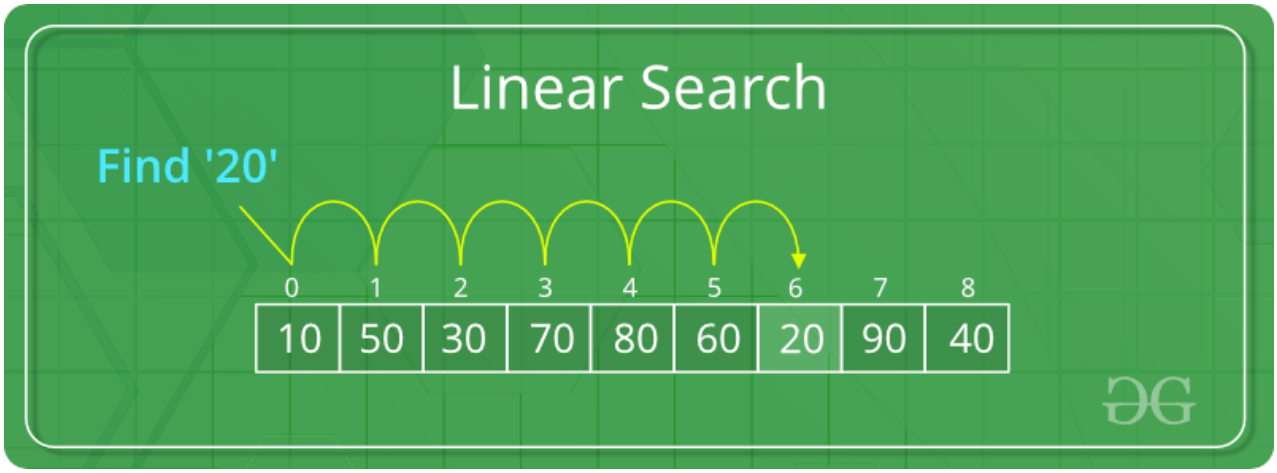
4. Sequential compare each element with proceeding one e.g. A[n]>A[n-1]
5. If the A[n] is greater than A[n-1] swap A[n-1] to A[n] and A[n-1] to A[n] with the help of variable K. Otherwise do nothing
6. Repeat step 3 to 5 until all elements in the array are sorted

A Program to implement bubble sort

```
package BICT_DAY;
public class Main
{
    public static void main(String args[])
    {
        int A[]=new int[]{16,10,4,6,2};
        for(int i=0;i<=4;i++)
        {
            for(int j=0;j<=3;j++)
            {
                int K;
                K=A[j];
                if(A[j]>A[j+1])
                {
                   A[j]=A[j+1];
                   A[j+1]=K;
                }
            }
        }
        System.out.println("Sorted Array is");
        for(int k=0;k<=4;k++)
        {
           System.out.println(A[k]);
        }
    }
}
```

Sequential SearchAlgorithm Using Array

Linear Search is defined as a sequential search algorithm that starts at one end and goes through each element of a list until the desired element is found, otherwise the search continues till the end of the data set. It is the easiest searching algorithm.



Sequential search algorithm

1. Declare an Array ‘A’
2. Store elements in the Array ‘A’
3. Declare variable K and store the number to be searched
4. Sequential compare each element in the array with variable K e.g. A[n]= =K, A[n+1] = =K, A[n+2] = =K until the element found
5. Return 1 if the element found Otherwise return -1
6. Display to the use the element was found

Example a Java Program to search number 20 in the group of numbers

```java
package BICT_DAY;
public class Main
{
    static int findvalue(int A[],int K)
    {
        for(int i=0;i< A.length;i++)
        {
            if(A[i]==K)
                return 1;
        }
        return -1;
    }
    public static void main(String args[])
    {
        int A[]=new int[]{10, 50, 30, 70, 80,60,20,90,40};
        int K=20;
        if(findvalue(A,K)==1)
        {
            System.out.println("The element Found");
        }
        else
        {
            System.out.println("The element not Found");
        }
    }
}
```

Radix Sort Algorithm

Radix sort is the linear sorting algorithm that is used for integers. In Radix sort, there is digit by digit sorting is performed that is started from the least significant digit to the most significant digit. Radix sort is a sorting algorithm that sorts the elements by first grouping the individual digits of the same **place value**. Then, sort the elements according to their increasing/decreasing order.

Suppose, we have an array of 10 elements. First, we will sort elements based on the value of the unit place. Then, we will sort elements based on the value of the tenth place. This process goes on until the last significant place.

The Radix sort algorithm

1. Define 10 queues each representing a bucket for each digit from 0 to 9.
2. Consider the least significant digit of each number in the list which is to be sorted.
3. Insert each number into their respective queue based on the least significant digit.
4. Group all the numbers from queue 0 to queue 9 in the order they have inserted into their respective queues.
5. Repeat from step 3 based on the next least significant digit.
6. Repeat from step 2 until all the numbers are grouped based on the most significant digit.

Example: Sorting the following list 132, 543, 783, 063, 007, 898, 049

Step 1: Consider the numbers at once space

13**2**, 54**3**, 78**3**, 06**3**, 00**7**, 89**8**, 04**9**

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
		13**2**	54**3**				00**7**	89**8**	04**9**
			78**3**						
			06**3**						

Step 2: Consider the numbers at tens space

1**3**2, 5**4**3, 7**8**3, 0**6**3, 0**0**7, 8**9**8, 0**4**9

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
0**0**7			1**3**2	5**4**3		0**6**3		7**8**3	8**9**8
				0**4**9					

Step 2: Consider the numbers at hundreds space

007, **1**32, **5**43,**0**49, **0**63, **7**83,**8**98

0	1	2	3	4	5	6	7	8	9
007	132				543		783	898	
049									
063									

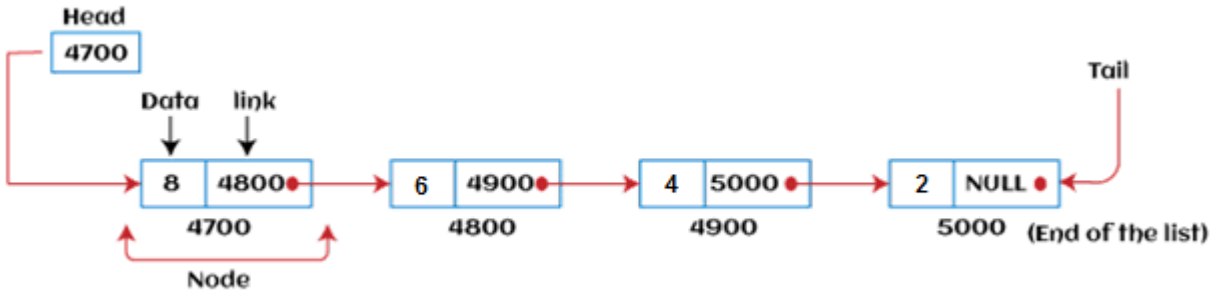
The Sorted array is 007,049,063,132,543,783,898

Linked list

Linked list is a linear data structure that includes a series of connected nodes. Linked list can be defined as the nodes that are randomly stored in the memory. A node in the linked list contains two parts, i.e., first is the data part and second is the address part. The last node of the list contains a pointer to the null. After array, linked list is the second most used data structure. In a linked list, every link contains a connection to another link.

Representation of a Linked list

Linked list can be represented as the connection of nodes in which each node points to the next node of the list. The representation of the linked list is shown below -



Till now, we have been using array data structure to organize the group of elements that are to be stored individually in the memory. However, Array has several advantages and disadvantages that must be known to decide the data structure that will be used throughout the program.

Why use linked list over array?

Linked list is a data structure that overcomes the limitations of arrays. Let's first see some of the limitations of arrays -

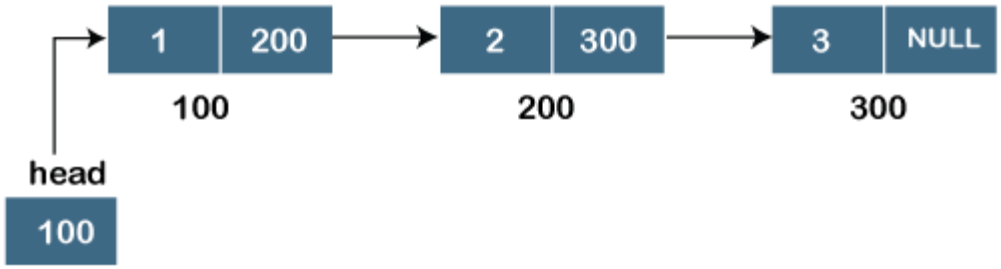
- The size of the array must be known in advance before using it in the program.
- Increasing the size of the array is a time taking process. It is almost impossible to expand the size of the array at run time.
- All the elements in the array need to be contiguously stored in the memory. Inserting an element in the array needs shifting of all its predecessors.

Linked list is useful because -

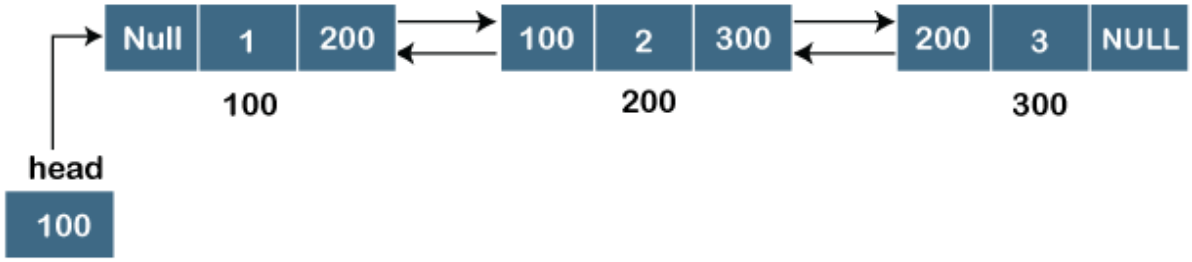
- It allocates the memory dynamically. All the nodes of the linked list are non-contiguously stored in the memory and linked together with the help of pointers.
- In linked list, size is no longer a problem since we do not need to define its size at the time of declaration. List grows as per the program's demand and limited to the available memory space.

Types of Linked list

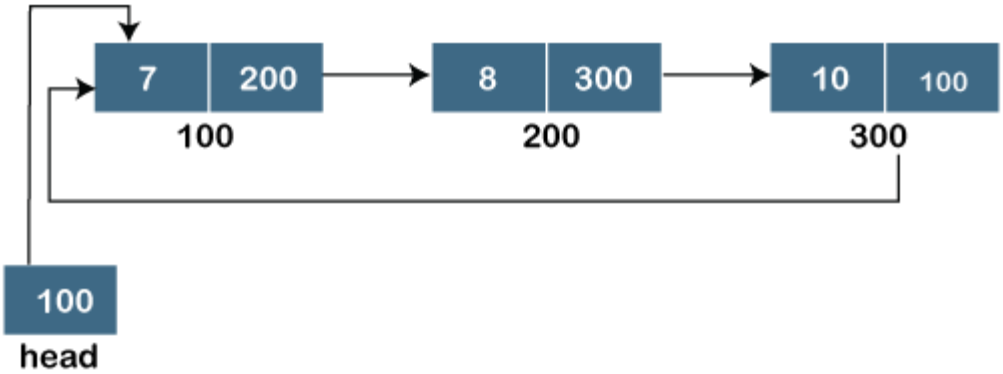
- **Singly-linked list -** Singly linked list can be defined as the collection of an ordered set of elements. A node in the singly linked list consists of two parts: data part and link part. Data part of the node stores actual information that is to be represented by the node, while the link part of the node stores the address of its immediate successor.



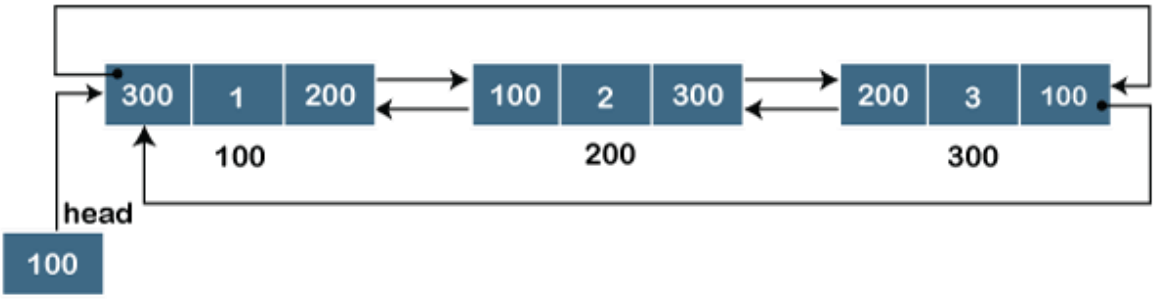
- **Doubly linked list -** Doubly linked list is a complex type of linked list in which a node contains a pointer to the previous as well as the next node in the sequence. Therefore, in a doubly-linked list, a node consists of three parts: node data, pointer to the next node in sequence (next pointer), and pointer to the previous node (previous pointer).



- **Circular singly linked list -** In a circular singly linked list, the last node of the list contains a pointer to the first node of the list. We can have circular singly linked list as well as circular doubly linked list.



- **Circular doubly linked list -** Circular doubly linked list is a more complex type of data structure in which a node contains pointers to its previous node as well as the next node. Circular doubly linked list doesn't contain NULL in any of the nodes. The last node of the list contains the address of the first node of the list. The first node of the list also contains the address of the last node in its previous pointer.



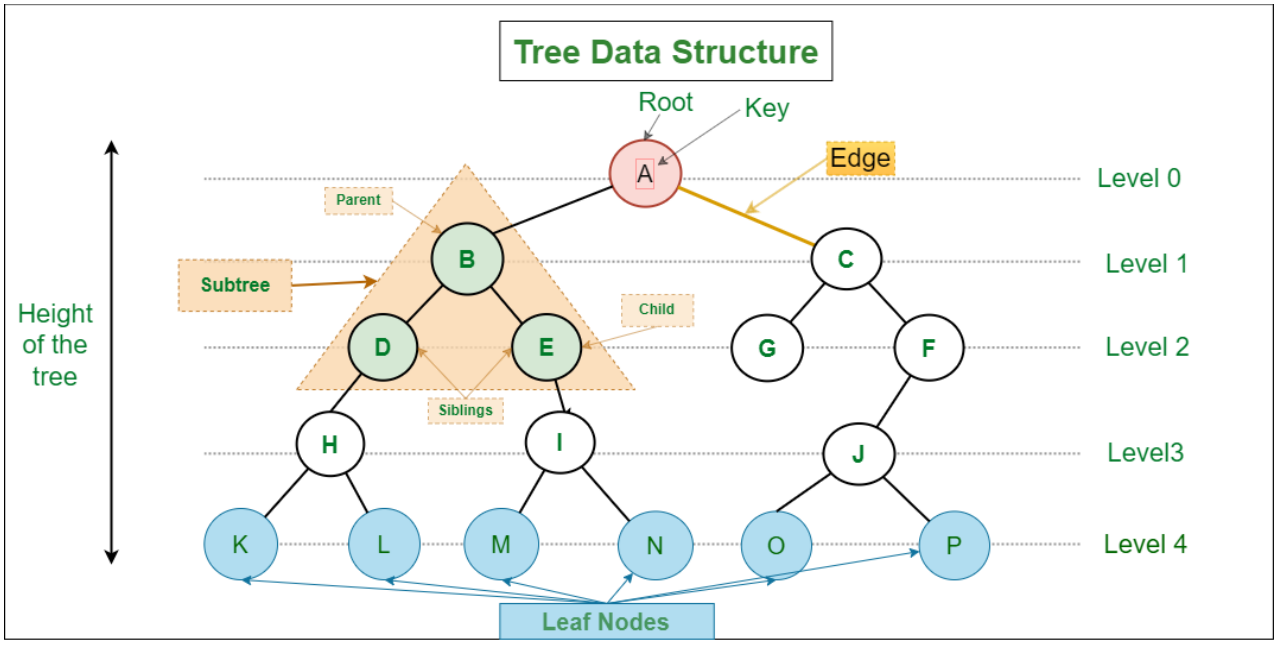
Operations performed on the Linked list

The basic operations that are supported by a list are mentioned as follows -

1. **Insertion** - This operation is performed to add an element to the list.
2. **Deletion -** It is performed to delete an operation from the list.
3. **Display** - It is performed to display the elements of the list.
4. **Search** - It is performed to search an element from the list using the given key.

Tree Data Structure

A tree data structure is non-linear because it does not store data sequentially. In a tree data structure, data (Elements) are stored in the nodes organized in a hierarchy. It is a hierarchical structure, as elements in a Tree are arranged on multiple levels. In the tree data structure, you cannot navigate to the child node without pathing the parent node



Basic Terminologies In Tree Data Structure:

i. **Parent Node:** The node which is a predecessor of a node is called the parent node of that node. **{B}** is the parent node of **{D, E}**.

ii. **Child Node:** The node which is the immediate successor of a node is called the child node of that node. Examples: **{D, E}** are the child nodes of **{B}.**

iii. **Root Node:** The topmost node of a tree or the node which does not have any parent node is called the root node. {A**}** is the root node of the tree. A non-empty tree must contain exactly one root node and exactly one path from the root to all other nodes of the tree.

iv. **Leaf Node or External Node:** The nodes which do not have any child nodes are called leaf nodes. **{K, L, M, N, O, P}** are the leaf nodes of the tree.

v. **Ancestor of a Node:** Any predecessor nodes on the path of the root to that node are called Ancestors of that node. **{A,B}** are the ancestor nodes of the node **{E}**

vi. **Descendant:** Any successor node on the path from the node to that leaf. **{E,I,M,N}** are the descendants of the node **{B}.**

vii. **Siblings:** Children of the same parent node are called siblings. **{D,E}** are called siblings.

viii. **Internal node:** A node with at least one child is called Internal Node.

ix. **Subtree**: Any node of the tree along with its descendant.

Properties of a Tree:

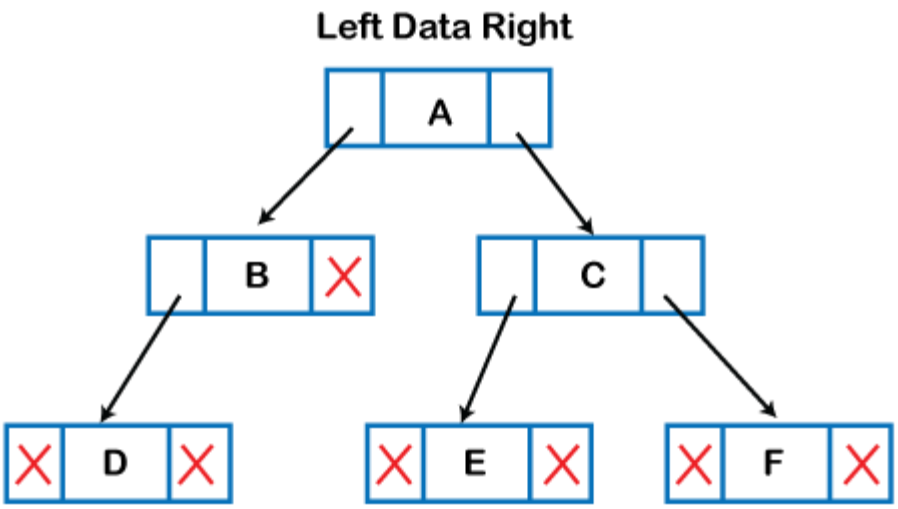
i. **Number of edges:** An edge can be defined as the connection between two nodes. If a tree has N nodes then it will have (N-1) edges. There is only one path from each node to any other node of the tree.

ii. **Depth of a node:** The depth of a node is defined as the length of the path from the node to the root node. Each edge adds 1 unit of length to the path. So, it can also be defined as the.Therefore the depth of root nodes is 0. E.g. Depth of Node **H** is **3**

iii. **Height of a node:** The height of a node can be defined as the length of the longest path from the node to the leaf of the tree. E.g. Height of **H** is **1,** height of **K** is **0, height of A(**height of a tree**)** is **4**

Implementation of Tree

The tree data structure can be created by creating the nodes dynamically with the help of the pointers. The tree in the memory can be represented as shown below:



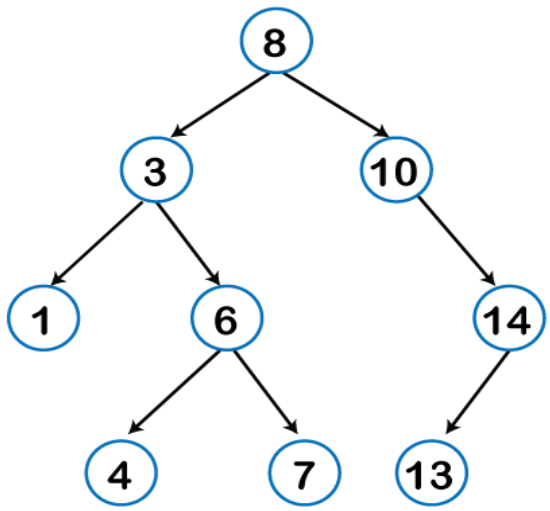
The above figure shows the representation of the tree data structure in the memory. In the above structure, the node contains three fields. The second field stores the data; the first field stores the address of the left child, and the third field stores the address of the right child.

Applications of trees

i. Storing naturally hierarchical data: Trees are used to store the data in the hierarchical structure. For example, the file system. The file system stored on the disc drive, the file and folder are in the form of the naturally hierarchical data and stored in the form of trees.
ii. Routing table: Tree data structures can be used to represent the topology of a network and to calculate routing tables for efficient data transmission.
iii. For information indexing in relational database.
iv. In websites to create XML and HTML documents
v. Binary Search Tree is a tree that allows fast search, insert, delete on a sorted data. It also allows finding closest item
vi. Spanning Trees and shortest path trees are used in routers and bridges respectively in computer networks

Binary tree

Binary name itself suggests two numbers, i.e., 0 and 1. In a binary tree, each node in a tree can have utmost two child nodes. Here, utmost means whether the node has 0 nodes, 1 node or 2 nodes.



Properties of Binary Tree

- At each level of i, the maximum number of nodes is 2^i.
- The height of the tree is defined as the longest path from the root node to the leaf node. The tree which is shown above has a height equal to 3. Therefore, the maximum number of nodes at height 3 is equal to (1+2+4+8) = 15. In general, the maximum number of nodes possible at height h is $(2^0 + 2^1 + 2^2 + \ldots 2^h) = 2^{h+1} - 1$.
- The minimum number of nodes possible at height h is equal to **h+1**.
- If the number of nodes is minimum, then the height of the tree would be maximum. Conversely, if the number of nodes is maximum, then the height of the tree would be minimum.

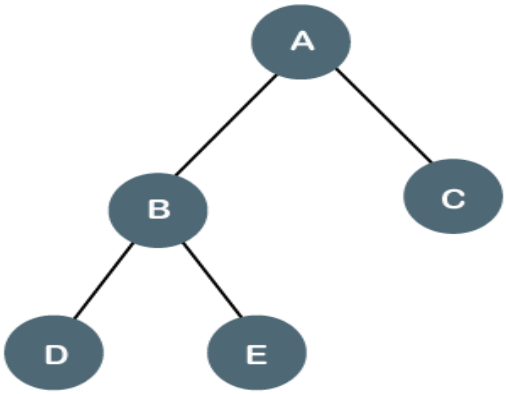
Types of Binary Tree

There are four types of Binary tree:

1. **Full/ proper/ strict Binary tree**

The full binary tree is also known as a strict binary tree. The tree can only be considered as the full binary tree if each node must contain either 0 or 2 children. The full binary tree can also be defined as the tree in which each node must contain 2 children except the leaf nodes.

Example:

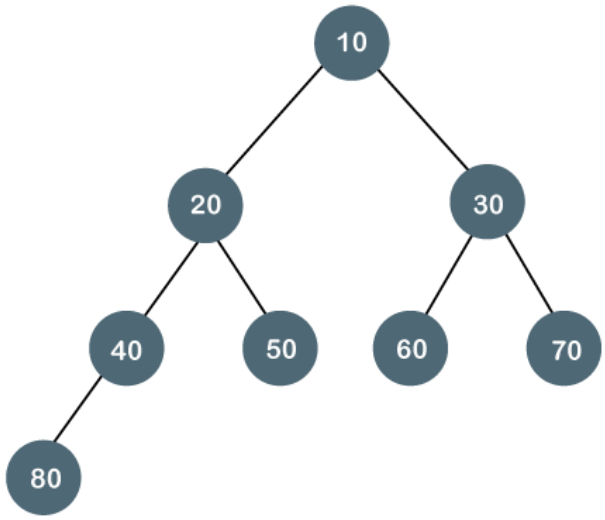


2. **Complete Binary Tree**

The complete binary tree is a tree in which all the nodes are completely filled except the last level. In the last level, all the nodes must be as left as possible. In a complete binary tree, the nodes should be added from the left.

The above tree is a complete binary tree because all the nodes are completely filled, and all the nodes in the last level are added at the left first.

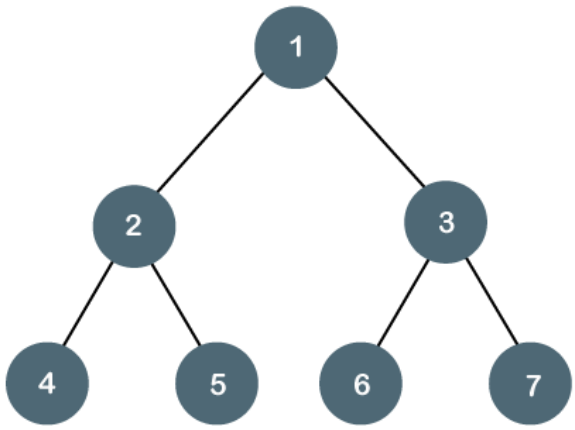
Example:



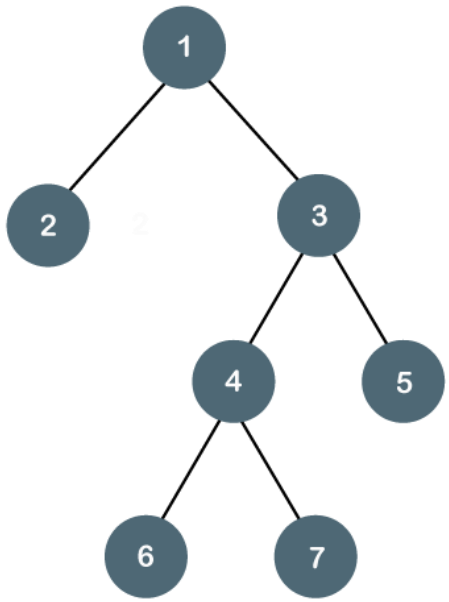
3. Perfect Binary Tree

A tree is a perfect binary tree if all the internal nodes have 2 children, and all the leaf nodes are at the same level.

Example:



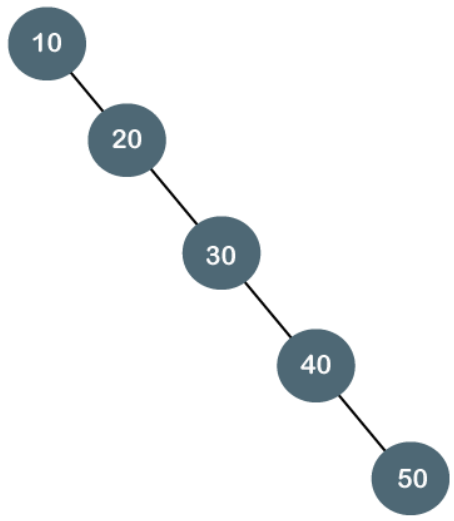
The below tree is not a perfect binary tree because all the leaf nodes are not at the same level.



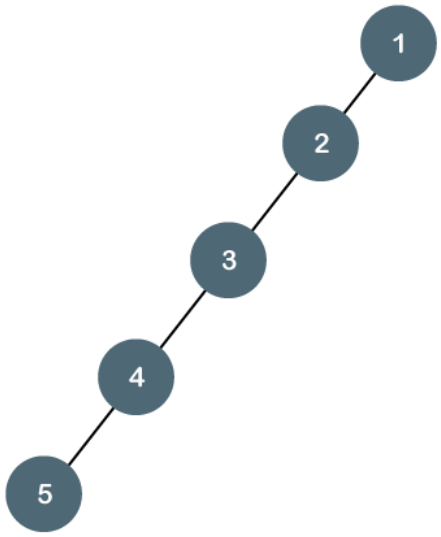
4. Degenerate Binary Tree

The degenerate binary tree is a tree in which all the internal nodes have only one child.

Example.



The above tree is a degenerate binary tree because all the nodes have only one child. It is also known as a right-skewed tree as all the nodes have a right child only.

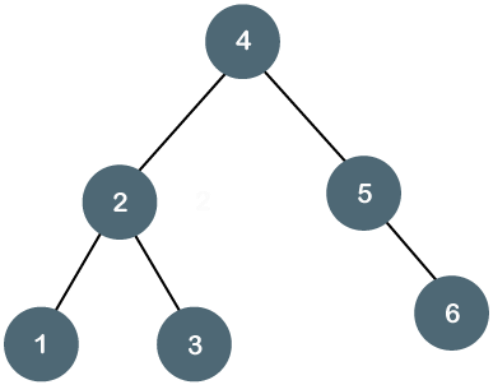


The above tree is also a degenerate binary tree because all the nodes have only one child. It is also known as a left-skewed tree as all the nodes have a left child only.

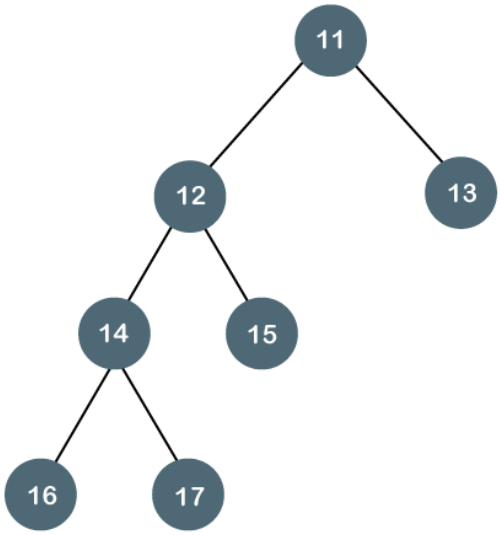
Balanced Binary Tree

The balanced binary tree is a tree in which both the left and right trees differ by atmost 1. For example, ***AVL*** and ***Red-Black trees*** are balanced binary tree.

Examples



The above tree is a balanced binary tree because the difference between the left subtree and right subtree is zero.



The above tree is not a balanced binary tree because the difference between the left subtree and the right subtree is greater than 1.